

Cable BOQ — Technical Review & Corrections

Bill of Quantities verification report — Power (Electrical) discipline

Project	P-379 / SABI — Proposed B+G+8+R Commercial & Residential Tower
Reviewed by	Mr. B.K. Murali (technical / domain review)
Verified against	docs/p379-power-boq-industry.xlsx (sample dated 15 June 2026)
Report date	17 June 2026
System	DrawToBOQ Estimation Pipeline (drawtoboq.com)

1. Executive Summary

A technical review of the P-379 power cable take-off raised concerns that the cable quantities — and the lighting schedule — did not appear to follow a proper floor-by-floor, project-specific procedure. The concern was that data may have been carried over from another project rather than read from this project's own drawings.

To confirm this, the reviewed Excel file was opened and its bills were inspected line by line. The review findings were verified as correct. The principal error is in the SMDB-to-DB distribution cable, where repeating floor boards were entered at a quantity of 1 instead of being multiplied across every floor, leaving the total cable quantity significantly understated.

All three issues are addressed in the live production system; the reviewed file was an older sample produced before those corrections. The lighting issue and full project-specific verification additionally require a fresh reading (re-extraction) of the drawings — not only a re-generation.

2. The Agreed Take-off Methodology

The correct cable take-off procedure, as set out in the review, has three parts:

Part A — Mains: LV Panel to SMDB

- Start at the Ground Floor LV Panel and open the schematic / single-line drawing.
- List every SMDB the panel feeds, with its reference number (typically one SMDB per floor).
- Against each SMDB record both the cable length and the cable size (size taken from the schematic, e.g. "150 sq.mm", as a short label).
- Measure the ground-floor run, then add the floor height (about 5 m) for each floor climbed up the riser.

Part B — Distribution: SMDB to DB

- Each SMDB feeds DBs on its floor. Typical floors repeat, so one floor is calculated and multiplied across all typical floors (e.g. $\times 8$ or $\times 16$).
- A floor-wise schedule must be prepared — measure floor by floor, then total.

Part C — Lighting Fixtures

- Fixture types and naming must be read from this project's own drawing legend — never reused from another or a domestic project.
- Quantities must be broken up floor by floor and then totalled.

3. How the Verification Was Carried Out

The sample file the reviewer had graded was opened directly and its key bills were extracted and inspected:

- File: docs/p379-power-boq-industry.xlsx (file date 15 June 2026 — i.e. before the corrections).
- Building confirmed from the sheets: B + G + 8 + R (Basement, Ground, 8 typical floors, Roof).
- Bills inspected: Bill 3 (SMDBs), Bill 4 (Distribution Boards), Bill 5 (LV Cables), Bill 8 (Lighting Fixtures).

The findings below quote the actual line items and quantities found in that file.

4. Findings — Evidence From the File

Finding A — Mains cable (LV !' SMDB): quantity close, description generic

Bill 5, section 5.1 lists the main feeders. The per-floor build-up is present — length increases by about 4 m for each floor climbed — which is why this part was “almost matching” in the review:

Feeder (to floor)	Cable size	Length (m)
SMDB-1F	4C × 150 mm ²	25
SMDB-2F	4C × 150 mm ²	29
SMDB-3F	4C × 150 mm ²	33
SMDB-4F	4C × 150 mm ²	37
SMDB-5F ... 8F	4C × 150 mm ²	41 / 45 / 49 / 53

Issue: the cable size is buried inside a long, repeated specification string — e.g. “4C × 150 mm² XLPE / SWA / PVC 600/1000 V armoured ...” with a generic brand list (Ducab / NCC / Oman / Riyadh). There is no clean, separate cable-size field. This is the generic, non-project-specific description noted in the review.

Finding B — Distribution cable (SMDB !' DB): the principal error

In Bill 4, repeating floor boards were entered as a single line at quantity 1, with no multiplication across floors:

Item	Description	Qty shown	Qty expected
4.1.15	DB-T01 to DB-T15 (odd floors)	1	15 × 4 = 60
4.1.16	DB-T01 to DB-T15 (even floors)	1	15 × 4 = 60
4.1.17	DB-SHOP01 to DB-SHOP12	1	12

So roughly 120 typical-floor DBs were counted as 2 boards, and 12 shop DBs as 1. The matching SMDB-to-DB cables in Bill 5 (sections 5.3 + 5.4) total only about 21 runs for the entire building, where roughly 120 would be expected. This understates the distribution cable quantity heavily and is exactly the “multiplication not done” issue raised in the review.

Finding C — Lighting fixtures: no floor-wise breakdown

Bill 8 lists each fixture type as a single lump quantity for the whole tower, referenced to a drawing range (“P-103 ... P-108”) with generic catalogue naming and brand lists — no per-floor schedule and no fixture-type tags from this project’s legend:

Item	Fixture type	Qty (whole building)
8.1.1	Recessed LED downlight 12 W	480
8.1.2	Surface LED panel 600×600	60
8.1.3	Linear LED batten 1200 mm	220
8.1.4	Bulkhead LED IP65	90
8.1.6	Pendant decorative (provisional)	8

There is no “1F = x, 2F = y ... total” breakdown, confirming the lighting concern raised in the review.

5. Corrections Applied (Production System)

The following corrections are implemented in the live estimation code and run automatically when an estimate is generated through the production route:

5.1 Dedicated cable-size column

- Bill 5 now uses a 9-column layout with a separate “Cable Size” column; the standard specification text is printed once per section rather than on every line.
- Code: `src/lib/excel/dubai-industry-boq-xlsx.ts` (`buildBill5`).

5.2 Floor / range expansion (the main fix)

- Aggregated entries such as “DB-T01 to DB-T15 (odd floors)” are automatically expanded into one line per (floor, board). For P-379 this turns about 2 summary lines into roughly 120 individual runs.
- Floor sets (odd / even / typical / 1F–8F) and tag ranges are both recognised and expanded.
- Code: `expandAggregatedCables()` in `src/lib/electrical/derive-cable-paths.ts`.

5.3 Per-floor lighting from the drawing

- A lighting-fixtures extraction step reads the drawing legend per floor; Bill 8 then groups fixtures floor by floor, flagging provisional items.
- If the legend was not captured during extraction, the system falls back to a generic list — so a fresh scan is needed in that case.
- Code: `buildBill8()` plus the `lighting_fixtures` extraction (prompt + schema).

5.4 Duplicate typical-floor SMDB removal

- Where per-floor boards (SMDB-1F ... 8F) and a generic typical-floor stack both exist, the duplicate stack and its feeders are removed to avoid double counting.
- Code: `dedupeTypicalSmdb()` in `src/lib/electrical/derive-cable-paths.ts`.

5.5 Cable length grounded in floor height

- The extraction prompt computes feeder length as: ~4 m lead-in + (floor index × typical floor height) + 0.5 m termination — matching the review’s “add the floor height per floor” rule, using the actual floor height rather than a flat 5 m.
- Code: `gateway-prompts/DRAWTOBOQ_ELECTRICAL_EXTRACT.md` (Step 10).

These corrections are wired into the estimate, power-BOQ, and gate routes via `enrichElectricalResult()`, so they run on every live generation. A verification script (`scripts/verify-p379-improvements.ts`) drives the production code over the P-379 data and checks the results.

6. Why the Reviewed File Still Showed the Problems

The BOQ generator exists in two parallel copies:

Copy	Location	Has the fixes?
Production	<code>src/lib/excel/*.ts</code> (used by the live app)	Yes — calls <code>enrichElectricalResult()</code>
Offline sample	<code>scripts/lib/*.mjs</code> (writes the docs sample)	No — never updated; uses an embedded fixture

The reviewed file (`docs/p379-power-boq-industry.xlsx`, 15 June) was produced by the offline copy, before the corrections, which is why every issue raised was genuinely present in it. A supporting change (`src/lib/excel/bill-columns.ts`) makes the pricing/brand/reference post-processors read columns by header name, so Bill 5’s added “Cable Size” column does not corrupt them.

7. Will Regenerating Improve the Output?

Issue	Fixed by regenerating?	Why
Mains description (Finding A)	Yes (via live system)	Presentation fix in buildBill5 — re-renders cleanly.
SMDB!DB quantity (Finding B)	Yes (via live system)	expandAggregatedCables() expands ranges on generation.
Lighting floor-wise (Finding C)	Only if legend was read	Else it falls back to a generic list — needs a re-scan.
Values truly from this drawing	No	Regeneration re-renders stored data; it does not re-read drawings.
Old offline sample script	No	That copy has not been updated yet.

In short: regenerating a live bid through the production route corrects Findings A and B immediately. Lighting and confirmation of project-specific data require a fresh scan of the drawings, not just a re-generation.

8. Remaining Actions

- Regenerate the BOQ for this project through the live production route to obtain the corrected mains description and per-floor distribution quantities.
- Re-scan (re-extract) the drawings if the lighting schedule must be project-specific and floor-wise, or to confirm all values were read from this project's drawings.
- Port the offline sample script (scripts/lib/*.mjs) to match the production logic so the docs sample reflects the corrected output.
- Optionally confirm with the original preparer whether the figures were taken from this project or adapted from another — as recommended in the review.

9. Conclusion

The review findings are valid and have been verified directly against the file. The principal quantity error — SMDB-to-DB cable counted as 1 instead of per floor — is corrected in the live system, along with the cable-size presentation. The lighting schedule and full project-specific verification still depend on a fresh reading of the drawings. The file that was reviewed was an older sample produced before these corrections.